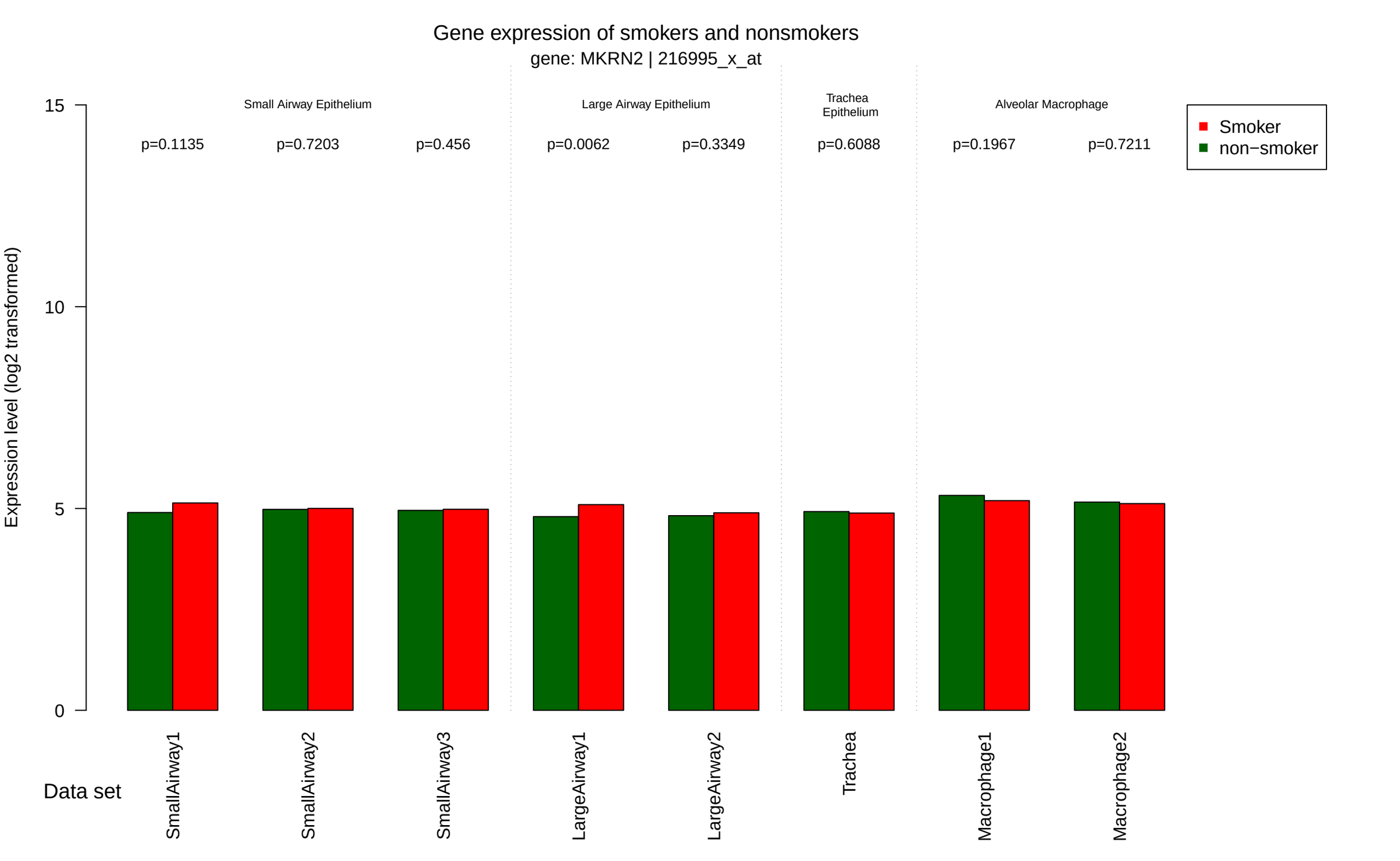
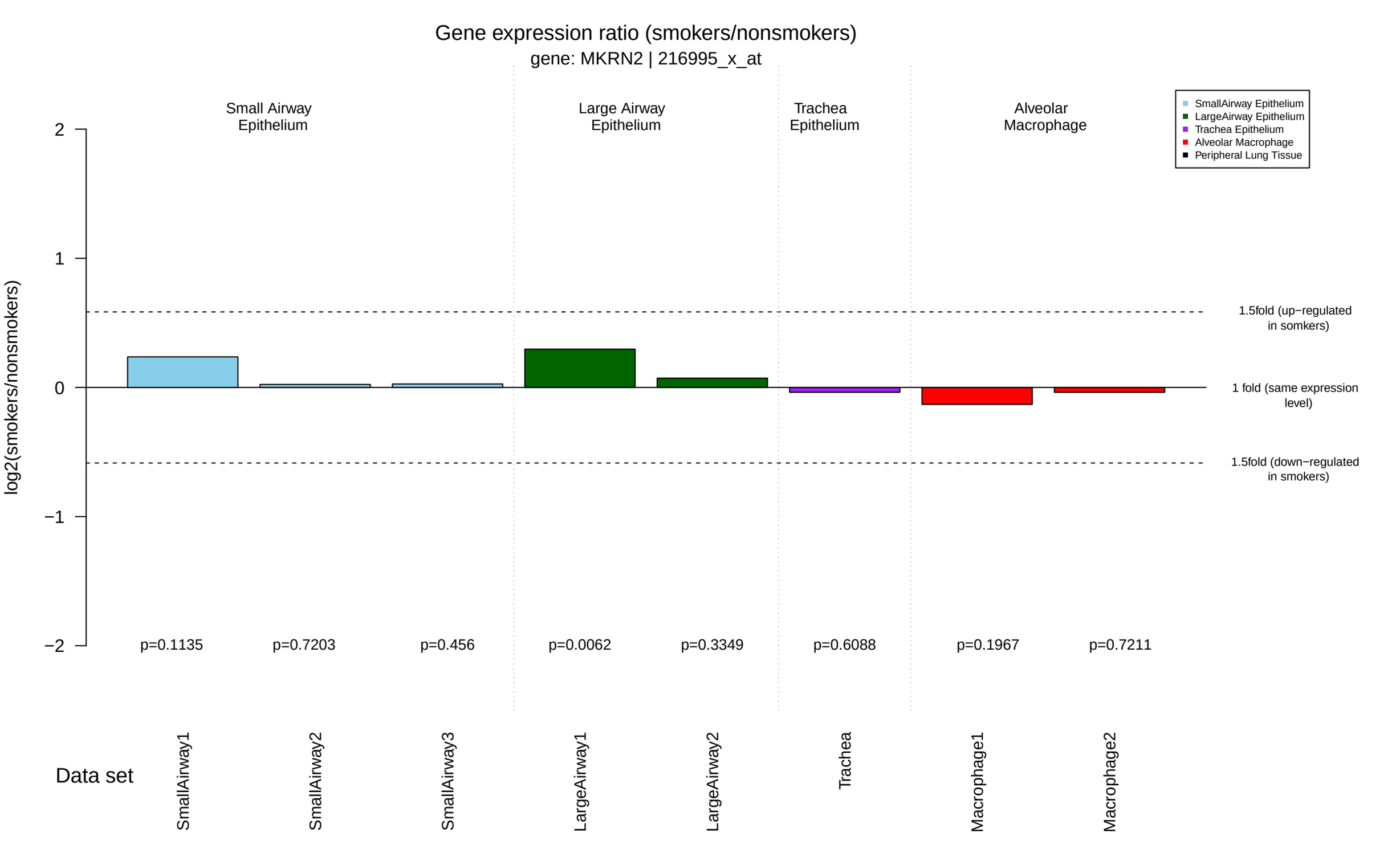
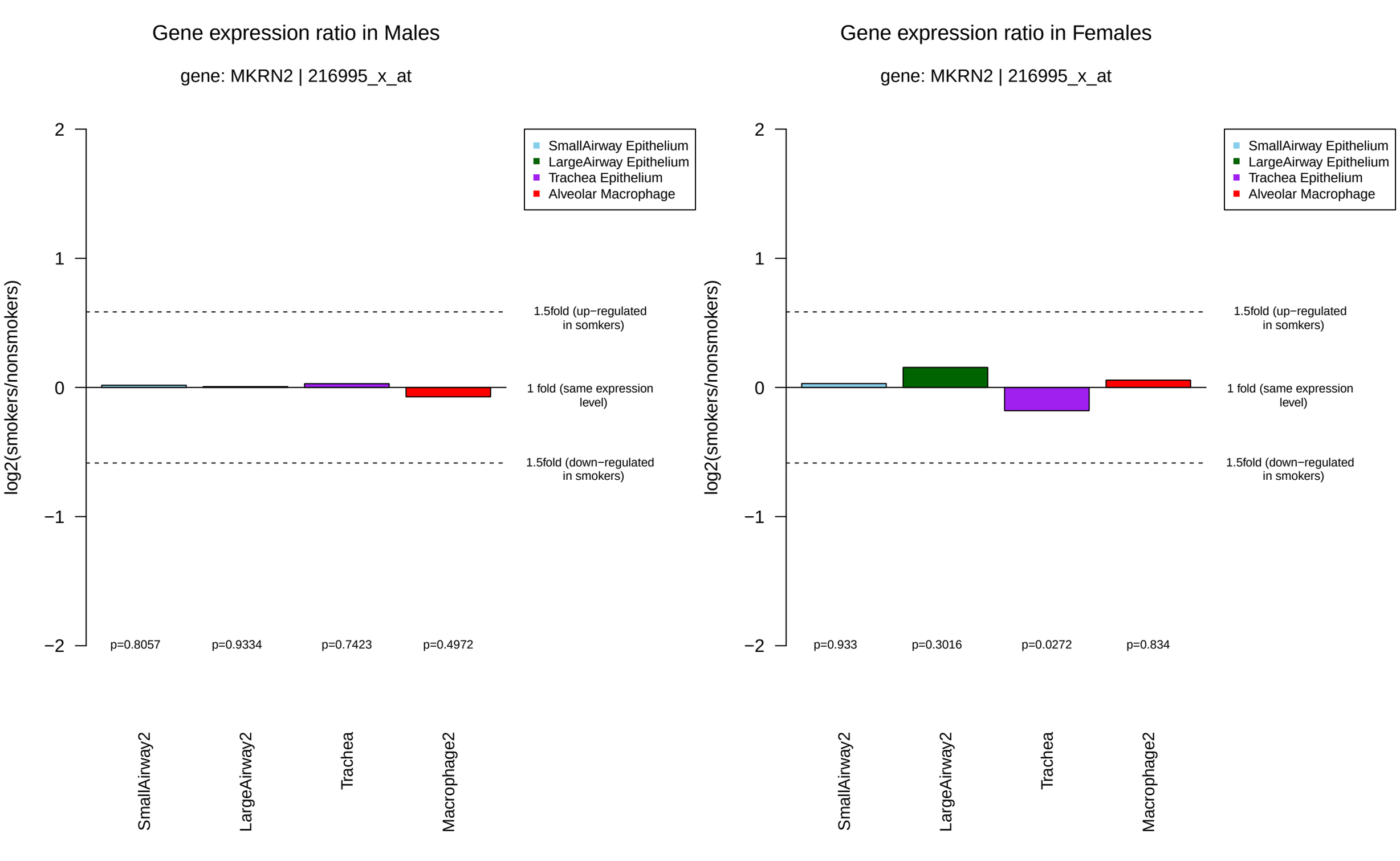
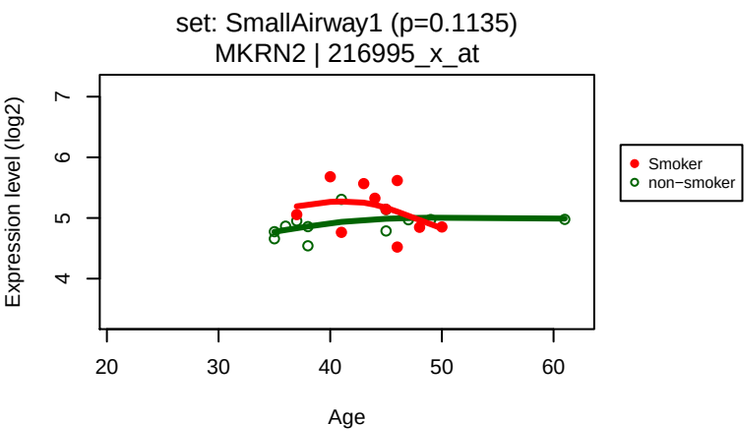
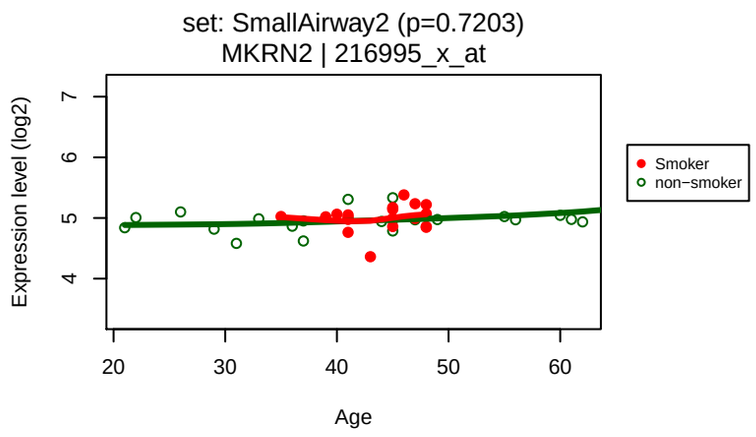
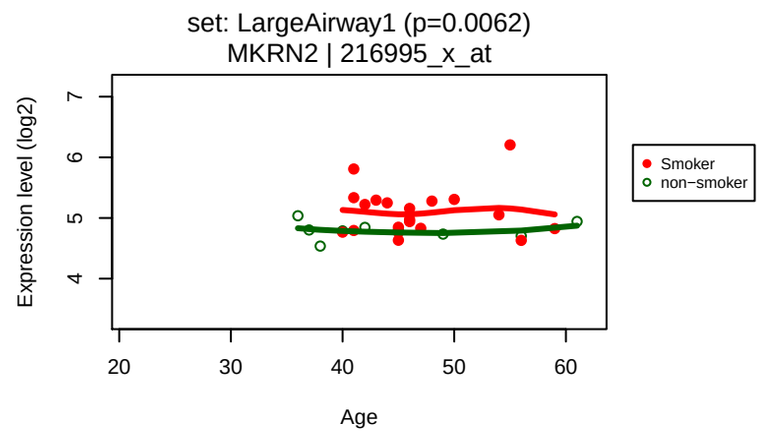

Gene expression of smokers and nonsmokers
gene: MKRN2 | 216995_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1135
p=0.7203
p=0.456
p=0.0062
p=0.3349
p=0.6088
p=0.1967
p=0.7211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

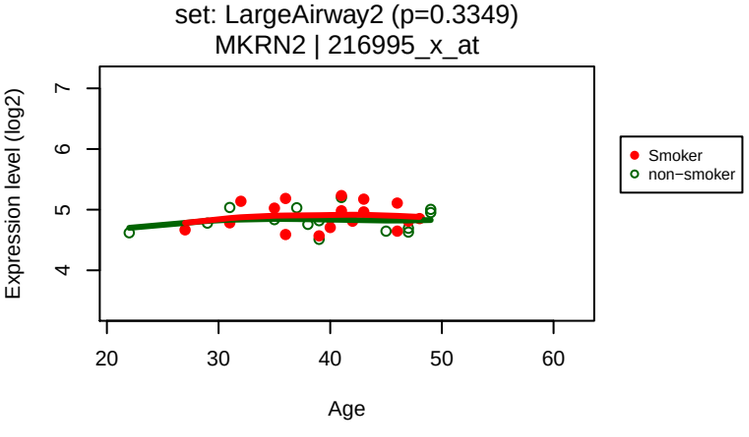

Gene expression ratio (smokers/nonsmokers)
gene: MKRN2 | 216995_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1135
p=0.7203
p=0.456
p=0.0062
p=0.3349
p=0.6088
p=0.1967
p=0.7211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

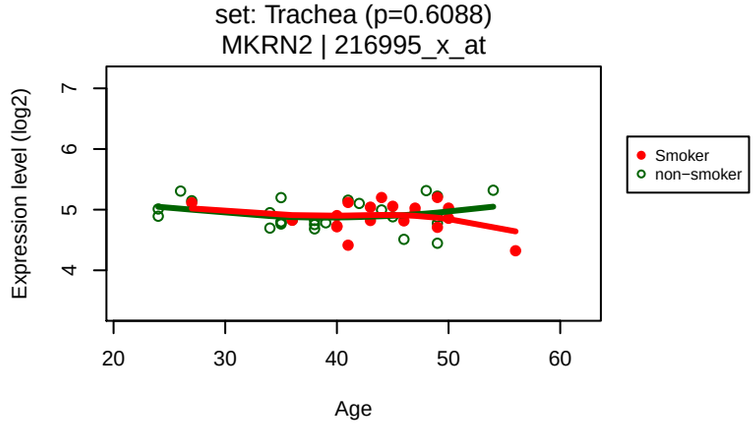

Gene expression ratio in Males
gene: MKRN2 | 216995_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8057
p=0.9334
p=0.7423
p=0.4972
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MKRN2 | 216995_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.933
p=0.3016
p=0.0272
p=0.834
SmallAirway2
LargeAirway2
Trachea
Macrophage2

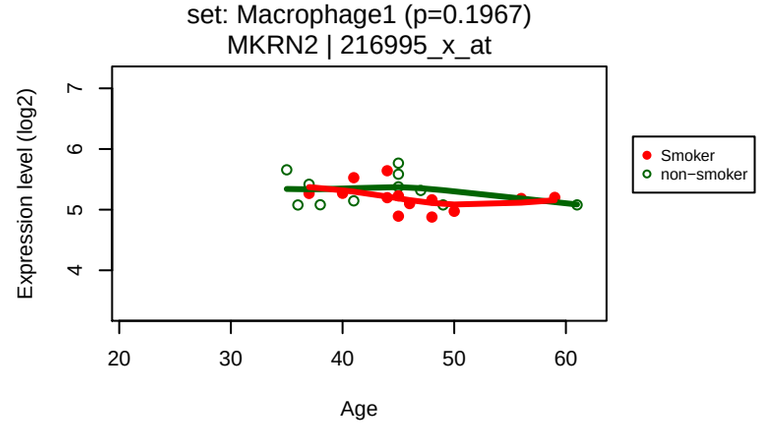

set: SmallAirway1 (p=0.1135)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker

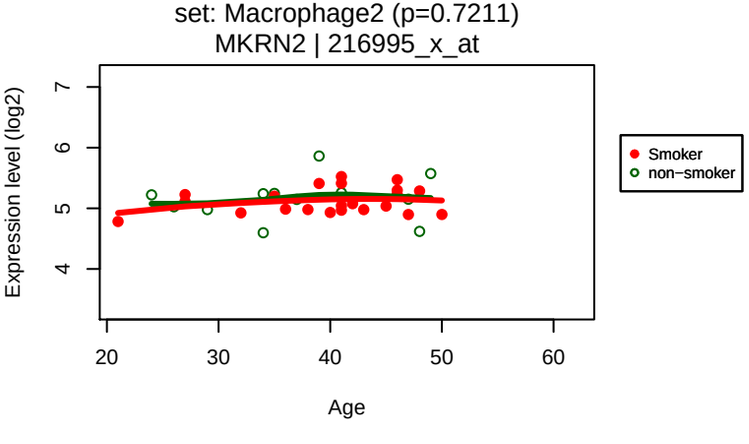

set: SmallAirway2 (p=0.7203)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker

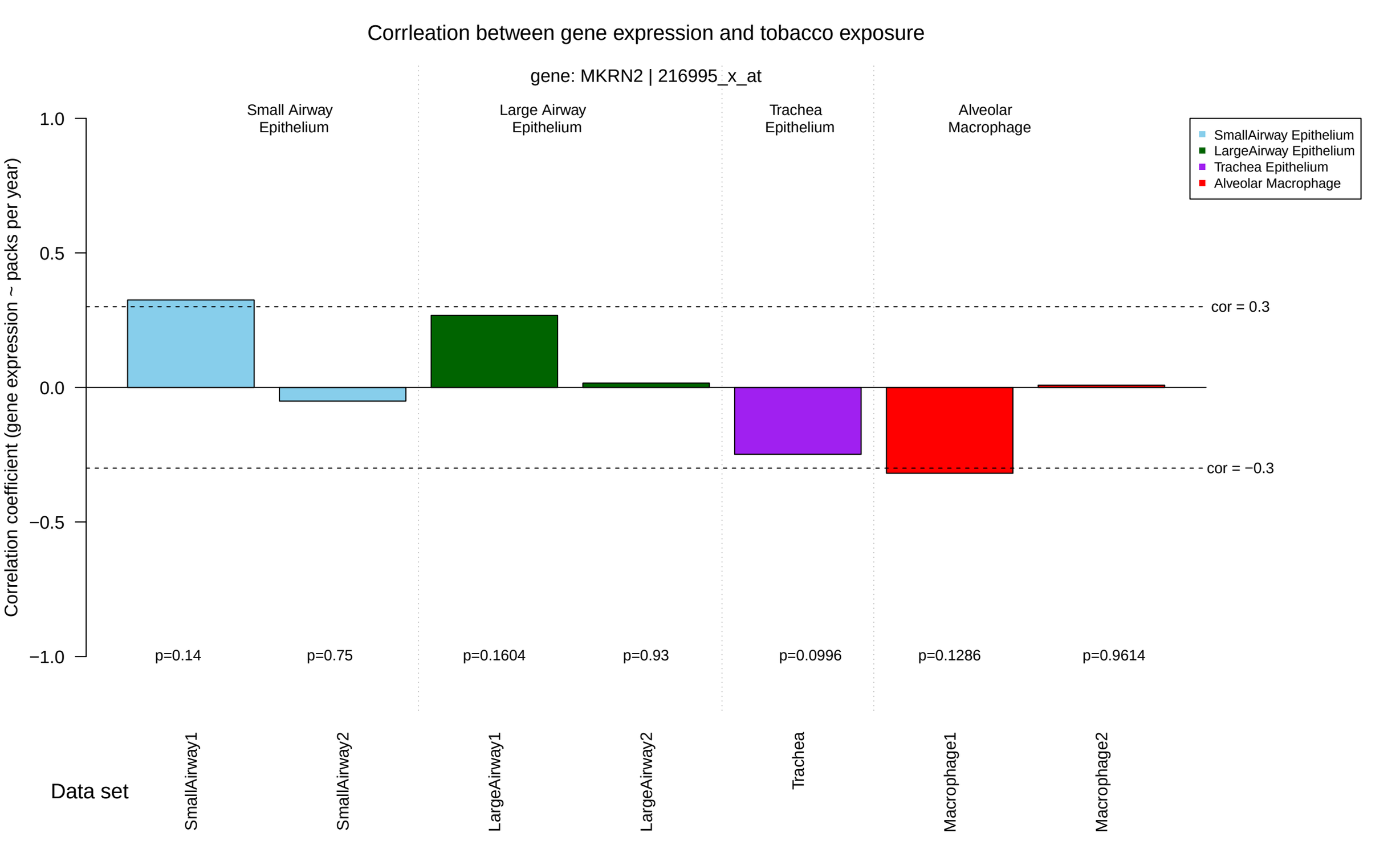

set: LargeAirway1 (p=0.0062)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3349)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6088)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1967)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7211)
MKRN2 | 216995_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MKRN2 | 216995_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.14
p=0.75
p=0.1604
p=0.93
p=0.0996
p=0.1286
p=0.9614
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2